

Onto Innovation (ONTO)

Initiate at Buy: AI packaging complexity drives process control growth, new products enable share gains

ONTO

12m Price Target: \$400.00

Price: \$331.71

Upside: 20.6%

We initiate coverage of ONTO with a Buy rating and a 12-month price target of \$400 (19% upside). We see a compelling risk/reward profile for ONTO as AI demand is underpinning idiosyncratic growth in inspection and metrology. As semiconductor designs grow in complexity and see increasing 3D integration, yield preservation has become a key focus. This structural shift dramatically increases process control intensity, positioning Onto's newly launched Dragonfly G5 and Atlas G6 platforms to capture market share. We expect Onto to outgrow the broader WFE market through 2028 while delivering strong margin expansion. **Our view is summarized by the following key points:**

- **We expect AI packaging complexity to drive process control outperformance.** The growing transition to advanced packaging multiplies required inspection steps and elevates wafer loss costs, making defect detection vital.
- **We believe Onto will outgrow WFE through 2028 as share gains could drive outperformance vs. peers.** The market has focused heavily on Onto's ability to recapture and sustain market share following prior share loss; we see its Dragonfly G5 driving share gains in the wafer-level packaging inspection market, aided by JetStep in the coming quarters.
- **We are confident in Onto's ability to drive margin upside through 2028.** Given its manufacturing transition to Asia and favorable product mix shift to next-generation tools, we expect margin expansion ahead of Street expectations, with 60% gross margins achievable in the medium term.
- **Our 12-month target price of \$400 reflects a 32X P/E multiple applied to our normalized EPS estimate of \$12.50.** Our target P/E multiple is near the high end of ONTO's historical range given strong growth in advanced packaging.

BUY

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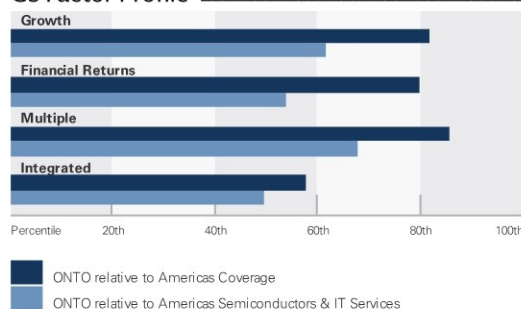
Key Data

Market cap: \$16.5bn
Enterprise value: \$16.6bn
3m ADTV: \$395.8mn
United States
Americas Semiconductors & IT Services
M&A Rank: 3

GS Forecast

	12/25	12/26E	12/27E	12/28E
Revenue (\$ mn)	1,005.3	1,453.3	2,009.1	2,467.3
EBITDA (\$ mn)	300.1	538.4	815.4	979.3
EBIT (\$ mn)	255.3	446.7	710.5	871.7
EPS (\$)	4.94	8.25	12.50	15.00
P/E (X)	26.4	40.2	26.5	22.1
EV/EBITDA (X)	20.2	30.8	19.8	15.9
FCF yield (%)	4.7	1.3	3.5	4.2
Dividend yield (%)	—	—	—	—
Net debt/EBITDA (X)	(1.2)	0.2	(0.6)	(1.2)
EPS (\$)	3/26	6/26E	9/26E	12/26E
	1.42	1.93	2.38	2.53

GS Factor Profile



Source: Company data, Goldman Sachs Research estimates.
See disclosures for details.

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BUY

Onto Innovation (ONTO)

Rating since Aug 17, 2026

Ratios & Valuation

	12/25	12/26E	12/27E	12/28E
P/E (X)	26.4	40.2	26.5	22.1
EV/EBITDA (X)	20.2	30.8	19.8	15.9
EV/sales (X)	6.0	11.4	8.0	6.3
FCF yield (%)	4.7	1.3	3.5	4.2
EV/DACF (X)	25.4	42.2	25.1	19.8
CROCI (%)	11.5	16.2	23.7	28.5
ROE (%)	12.1	19.6	26.7	25.9
Net debt/EBITDA (X)	(1.2)	0.2	(0.6)	(1.2)
Net debt/equity (%)	(16.5)	4.0	(18.8)	(36.7)
Interest cover (X)	NM	NM	NM	NM
Inventory days	211.3	200.7	191.1	169.0
Receivable days	104.8	85.6	84.6	80.6
Days payable outstanding	94.8	103.9	109.2	107.9

Growth & Margins (%)

	12/25	12/26E	12/27E	12/28E
Total revenue growth	1.8	44.6	38.2	22.8
EBITDA growth	(4.0)	79.4	51.4	20.1
EPS growth	(7.4)	67.0	51.5	19.9
DPS growth	NM	NM	NM	NM
Gross margin	49.7	54.1	58.4	59.0
EBIT margin	25.4	30.7	35.4	35.3

Price Performance

Source: FactSet. Price as of 14 Aug 2026 close.

Income Statement (\$ mn)

	12/25	12/26E	12/27E	12/28E
Total revenue	1,005.3	1,453.3	2,009.1	2,467.3
Cost of goods sold	(505.5)	(667.8)	(836.4)	(1,010.8)
SG&A	(234.9)	(329.0)	(372.0)	(439.0)
R&D	(132.0)	(160.5)	(206.4)	(260.6)
Other operating inc./exp.)	106.7	139.9	116.2	114.7
EBITDA	300.1	538.4	815.4	979.3
Depreciation & amortization	(60.4)	(102.5)	(104.9)	(107.6)
EBIT	255.3	446.7	710.5	871.7
Net interest inc./exp.)	35.0	19.3	11.0	4.0
Income/(loss) from associates	—	—	—	—
Pre-tax profit	285.3	465.1	721.5	875.7
Provision for taxes	(26.1)	(41.8)	(90.8)	(114.2)
Minority interest	—	—	—	—
Preferred dividends	—	—	—	—
Net inc. (pre-exceptionals)	243.5	412.6	630.7	761.6
Net inc. (post-exceptionals)	136.8	272.7	514.5	646.9
EPS (basic, pre-except) (\$)	4.96	8.31	12.60	15.09
EPS (diluted, pre-except) (\$)	4.94	8.25	12.50	15.00
EPS (ex-ESO exp., dil.) (\$)	--	--	--	--
DPS (\$)	—	—	—	—
Div. payout ratio (%)	0.0	0.0	0.0	0.0
Wtd avg shares out. (basic) (mn)	49.1	49.7	50.1	50.5
Wtd avg shares out. (diluted) (mn)	49.3	50.0	50.4	50.8

Balance Sheet (\$ mn)

	12/25	12/26E	12/27E	12/28E
Cash & cash equivalents	346.1	1,215.3	1,793.1	2,496.1
Accounts receivable	268.9	413.0	518.0	571.8
Inventory	298.3	436.0	439.8	496.3
Other current assets	354.7	672.1	672.1	672.1
Total current assets	1,268.0	2,736.4	3,423.1	4,236.4
Net PP&E	127.2	119.2	109.5	98.8
Net intangibles	942.1	902.2	902.2	902.2
Total investments	0.0	0.0	0.0	0.0
Other long-term assets	30.4	31.2	31.2	31.2
Total assets	2,367.7	3,789.0	4,465.9	5,268.5
Accounts payable	156.2	223.9	276.4	321.2
Short-term debt	—	—	—	—
Current lease liabilities	—	—	—	—
Other current liabilities	62.7	130.6	240.4	351.4
Total current liabilities	218.9	354.5	516.9	672.6
Long-term debt	—	1,300.0	1,300.0	1,300.0
Non-current lease liabilities	—	—	—	—
Other long-term liabilities	48.1	33.0	33.0	33.0
Total long-term liabilities	48.1	1,333.0	1,333.0	1,333.0
Total liabilities	267.1	1,687.5	1,849.9	2,005.6
Preferred shares	—	—	—	—
Total common equity	2,100.6	2,101.5	2,616.1	3,262.9
Minority interest	—	—	—	—
Total liabilities & equity	2,367.7	3,789.0	4,465.9	5,268.5
BVPS (\$)	42.77	42.31	52.26	64.66

Cash Flow (\$ mn)

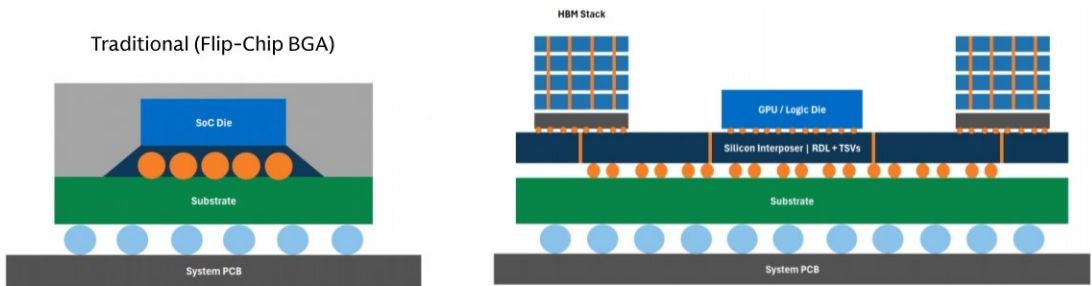
	12/25	12/26E	12/27E	12/28E
Net income	136.8	272.7	514.5	646.9
D&A add-back	88.1	133.8	135.9	139.8
Minority interest add-back	—	—	—	—
Net (inc)/dec working capital	60.1	(179.6)	(56.3)	(65.6)
Others	43.4	2.4	—	—
Cash flow from operations	328.3	229.3	594.1	721.0
Capital expenditures	(28.5)	(14.8)	(16.4)	(18.1)
Acquisitions	—	—	—	—
Divestitures	—	—	—	—
Others	(93.1)	(335.5)	—	—
Cash flow from investing	(121.6)	(350.4)	(16.4)	(18.1)
Dividends paid	—	—	—	—
Share issuance/(repurchase)	(61.6)	(199.1)	—	—
Inc/(dec) in debt	—	1,205.3	—	—
Others	(12.0)	(16.0)	—	—
Cash flow from financing	(73.6)	990.3	0.0	0.0
Total cash flow	133.2	869.2	577.8	702.9
Free cash flow	299.8	214.4	577.8	702.9
Free cash flow per share (basic) (\$)	6.10	4.32	11.54	13.93

Source: Company data, Goldman Sachs Research estimates.

We expect AI packaging complexity to drive process control outperformance

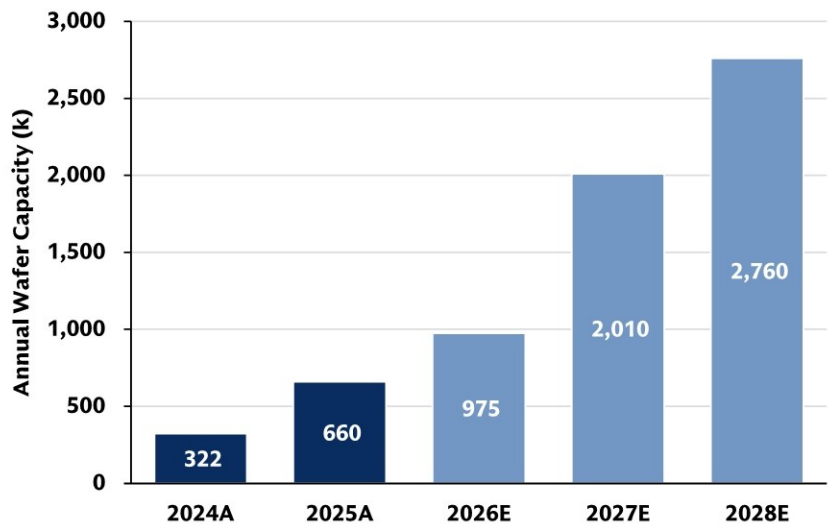
As the semiconductor industry reaches physical limits in shrinking transistors, chip designers are increasingly turning to advanced packaging chip architectures to continue improving performance over successive generations of process technology. This technology involves stacking multiple separately manufactured dies (chips) side-by-side (2.5D) or vertically (3D) to maximize the performance of a processor. High bandwidth memory (HBM) chips are an example of 3D packaging, where multiple (typically 8, 12, or 16 layers) of DRAM (dynamic random access memory) chips are stacked vertically and connected by through-silicon vias (TSVs), vertical holes in the stack filled with copper allowing signals to move throughout the stack. TSMC’s CoWoS (Chip-on-Wafer-on-Substrate) is a type of 2.5D packaging, in which multiple high value logic dies and HBM stacks are integrated side-by-side on a single interposer.

Exhibit 1: Comparison of traditional and 3D packaging (CoWoS)



Source: Goldman Sachs Global Investment Research

Exhibit 2: TSMC’s CoWoS capacity ramp

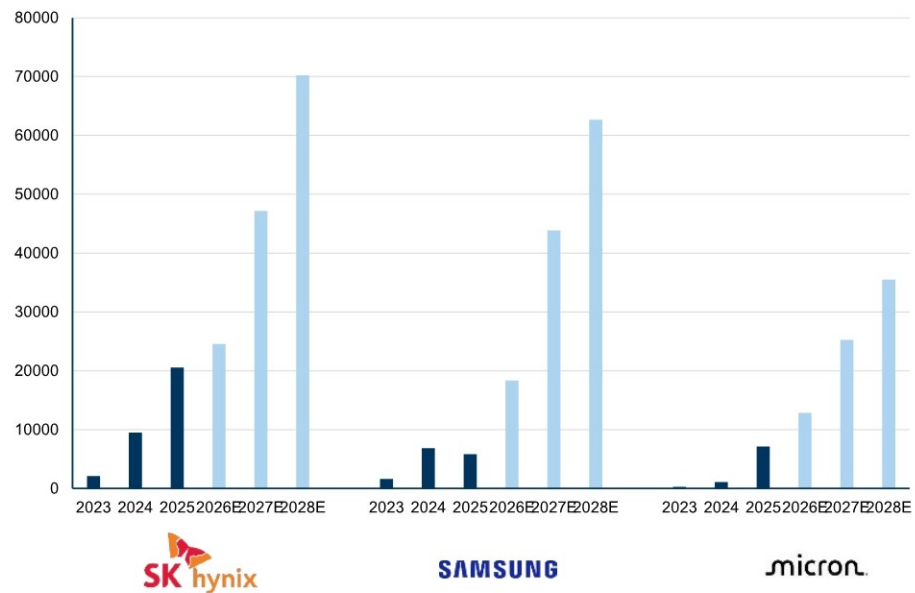


Source: Gartner

Our strong growth expectations for advanced packaging technology are directly reflected in our company-specific forecasts for leading chip manufacturers. We expect TSMC’s CoWoS capacity to expand at a 73% CAGR through 2028, driven by demand for the leading-edge AI accelerators utilizing this architecture, most notably NVIDIA’s Blackwell (B200/GB200), AMD’s MI350, and Google’s TPU v7/v8. Concurrently, we estimate the global HBM market, tightly controlled by SK Hynix, Samsung, and Micron,

will grow at a 71% CAGR through 2028, propelled by the exponential rise in memory capacity and bandwidth requirements per GPU.

Exhibit 3: Major DRAM suppliers have significantly ramped their HBM capacity

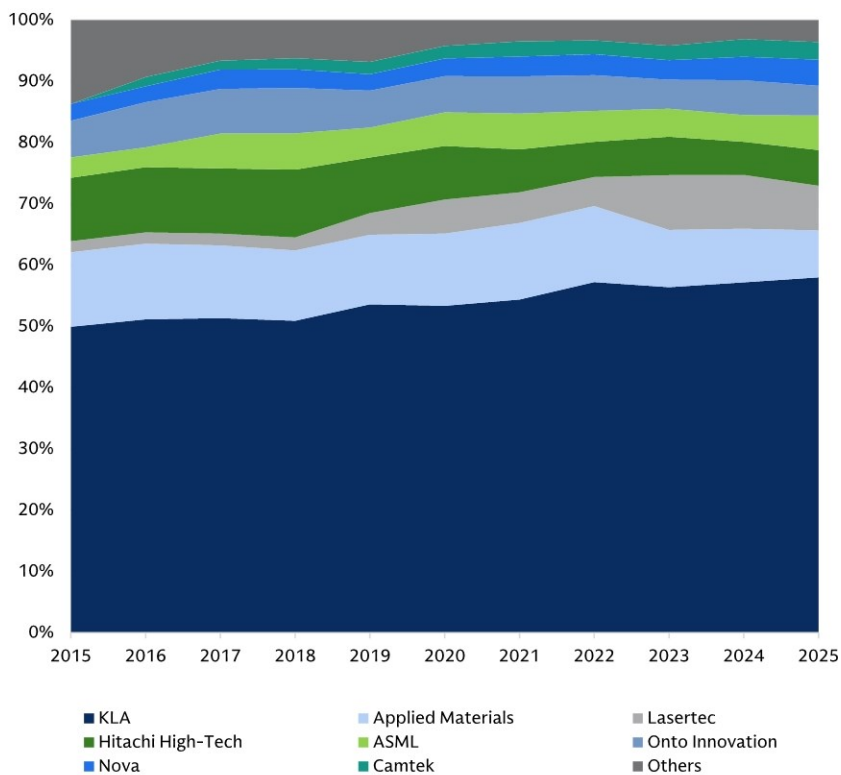


Source: Gartner

This transition to complex multi-chip architectures significantly increases the number of required process control steps to inspect micro-bumps, vias, and other features with sub-micron alignment. With an increasing number of high-cost chips in a single package, the financial risk of yield loss rises exponentially. In traditional packaging, a defect only ruins a single die, while a defect introduced in a multi-chip package can compromise an entire system containing multiple high-value logic and memory chips. This structure drives demand for process control equipment, which encompasses metrology (measuring microscopic physical features) and inspection (detecting defects or contamination).

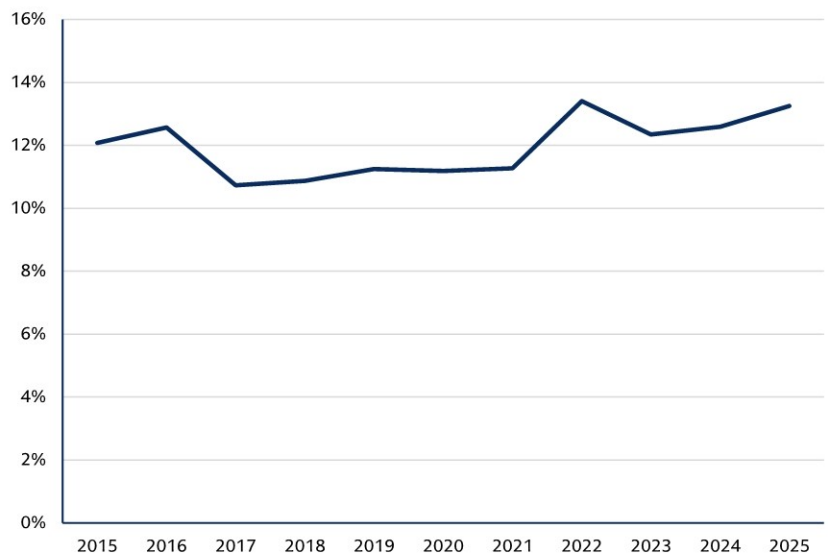
The overall semiconductor manufacturing process control market has grown at a 14% CAGR over the last 10 years, reaching ~\$15 bn in 2025 and accounting for approximately 13% of WFE spending. While the rest of the market is fragmented across more specialized players, KLA remains the dominant player at 58% share in 2025, driven by its entrenched position in front-end optical inspection. Onto captured roughly 4.8% in 2025 through its portfolio which spans both front-end and back-end applications. The company has a presence across the macro defect inspection, wafer-level packaging inspection, unpatterned wafer inspection, and thin film metrology markets.

Exhibit 4: Market shares of key industry players in process control, 2015-2025



Source: Gartner, Data compiled by Goldman Sachs Global Investment Research

Exhibit 5: Process control as a percentage of WFE, 2015-2025

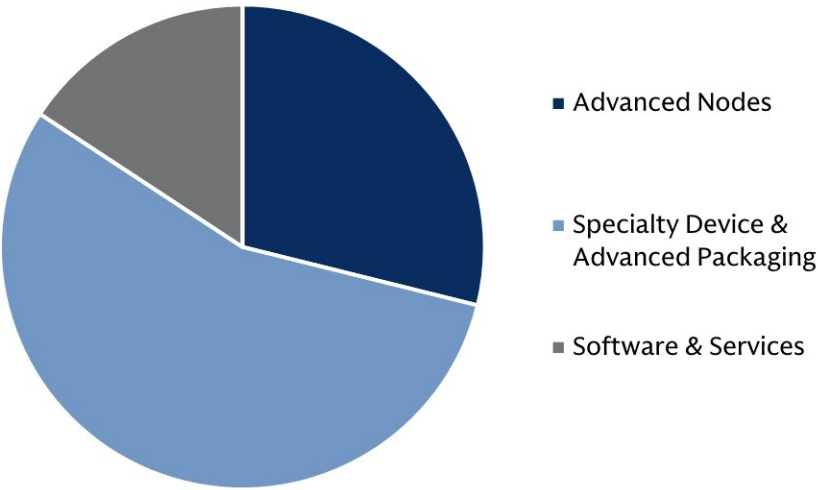


Source: Gartner

In the front end, thin film metrology is a notable market for Onto as the transition to gate-all-around (GAA) transistors drives demand for its Atlas G6 tool, supporting management’s expectations of approximately 25% advanced nodes growth in 2026. However, in recent years, Onto has established its strongest position in the wafer-level packaging inspection market, securing a 27% share position in 2025. This specialized

sector, which inspects intact wafers before dicing to detect defects and protect yields, was led by KLA Corporation at 48%, with Camtek holding 24%. Looking ahead, we expect Onto’s revenue mix to shift increasingly toward the specialty device and advanced packaging segment, which we estimate to grow 56% in 2026, reaching 55% of total revenue. We forecast advanced nodes and software and services at 29% and 16% of revenue, respectively.

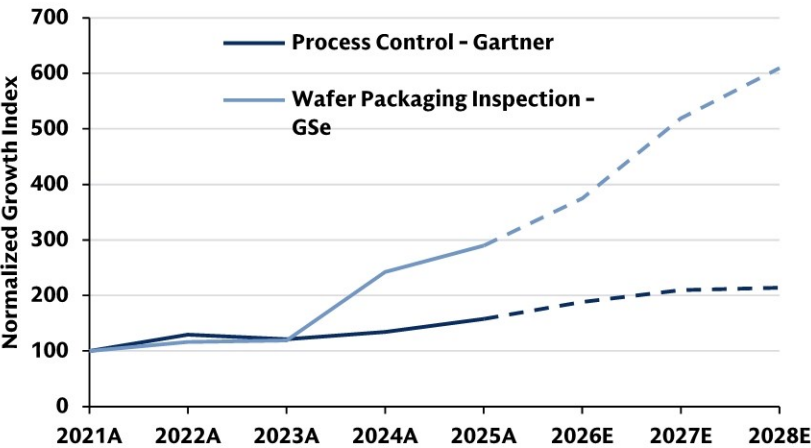
Exhibit 6: Onto revenue composition (2026E)



Source: Goldman Sachs Global Investment Research

Given the rapid adoption of advanced packaging, we project the wafer-level packaging inspection market to outpace both the broader process control market and overall WFE, growing at a CAGR of 31% through 2028, creating a highly favorable demand backdrop for Onto to leverage its Dragonfly G5 platform targeting this market.

Exhibit 7: Wafer-level inspection market growth relative to the process control market (indexed to 100 in 2021 baseline)



Source: Gartner, Goldman Sachs Global Investment Research

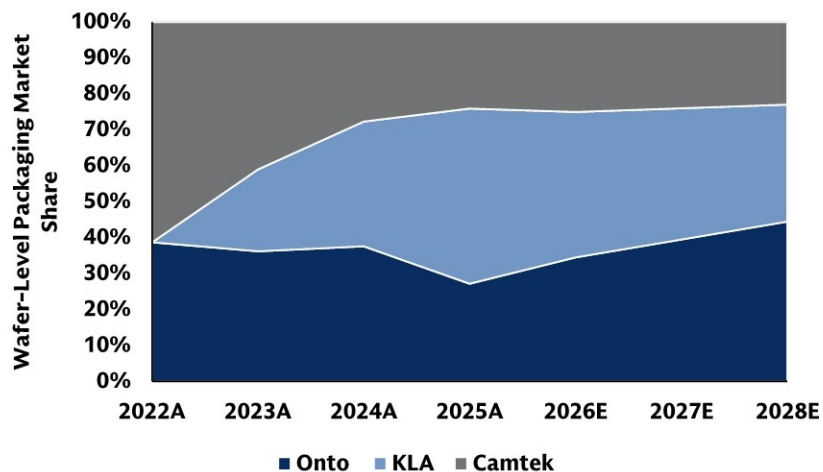
The authors would like to thank Ruth Pietrzykowski for her contribution to this publication.

Onto's next-generation products should drive market share gains

Although TSMC has been one of Onto's largest long-term customers, Onto lost significant share in late 2024/early 2025 following TSMC's large-scale production ramp of CoWoS-L. Onto's share in the wafer-level packaging inspection market fell from 37% (2024) to 27% (2025). At the time, the Dragonfly G3 was Onto's most advanced system for advanced packaging applications. However, its precision (or sensitivity) did not meet the requirements TSMC needed for CoWoS-L, opening the door for competitors to gain ground at one of Onto's largest advanced packaging customers.

Management accelerated development of Onto's next-generation inspection platform, launching Dragonfly G5 in March 2026 and securing qualification from TSMC's New Tool Selection Committee for 2.5D advanced packaging in April 2026. The Dragonfly G5 platform delivers sub-micron 2D/3D macro defect inspection with sensitivity down to 150 nm and throughput up to 5X faster than the G3. The platform's validation was reinforced by a \$240 million volume purchase agreement (VPA) with a leading HBM manufacturer through 2027, supporting our view that Onto is regaining momentum and can recapture share from peers KLA and Camtek. We estimate Onto's wafer-level packaging inspection share will rise from 27% in 2025 to 42% by 2028.

Exhibit 8: Onto's share of wafer-level packaging market vs. Camtek and KLA, 2022-2028



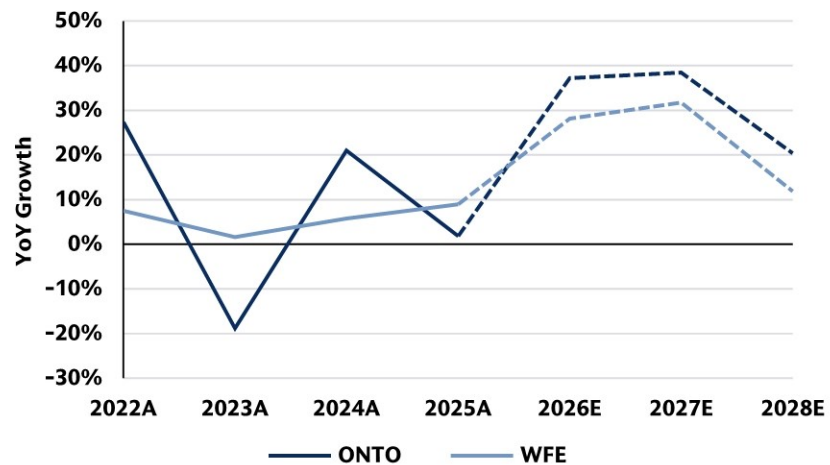
Source: Goldman Sachs Global Investment Research

We believe Onto's ability to defend and grow market share rests on a structural tradeoff in the industry between precision and throughput. Although KLA is a dominant player in high-resolution inspection, we believe its higher platform cost, adapted from its near-dominance in front-end applications, limits its cost of ownership advantage in the back-end applications. Onto leverages its deep back-end experience to offer a throughput and cost-of-ownership edge that becomes highly compelling as customers transition to high-volume manufacturing.

The HBM VPA extends the share-gain potential beyond CoWoS into hybrid bonding, where Onto holds a distinct advantage in laser-based inspection. Hybrid bonding requires infrared (IR) subsurface inspection to detect buried voids, a capability the company inherited from Rudolph Technologies following its 2019 merger with Nanometrics to form Onto Innovation. While competitors such as Camtek rely primarily

on white-light triangulation and interferometry for surface metrology, Onto's laser-based illumination (such as its Clearfind technology) and integrated IR capabilities in the Dragonfly G5 allow it to detect organic residues and subsurface defects that traditional white-light optics can miss. This integrated, multi-mode optical system provides Onto with a strong competitive position during the hybrid bonding ramp across major HBM players. We expect this market share recovery and advanced packaging momentum to drive Onto to outgrow the broader WFE market through 2028.

Exhibit 9: Onto's revenue growth vs. WFE spending, 2022-28

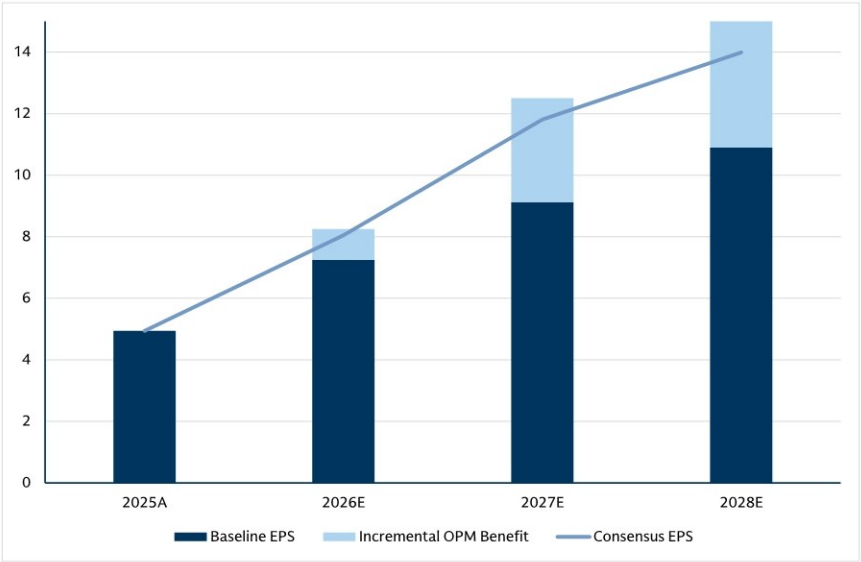


Source: Gartner, Goldman Sachs Global Investment Research

We expect Onto to drive margin upside through 2028

We believe Onto is positioned for significant operating margin expansion through 2028 as operating leverage should be complemented by a favorable product mix shift toward higher-value advanced packaging and next-generation metrology platforms. Additionally, the continued transition of tool shipments from extended factories in Asia should structurally lower manufacturing costs and improve supply-chain efficiency over time. Finally, we view the company's investment in Semilab as margin accretive, adding higher-value metrology capabilities that support a more profitable growth profile.

Exhibit 10: Incremental EPS benefit from higher operating margins
Baseline EPS reflects 2025A non-GAAP operating margin applied to forward GSe revenue



Source: Goldman Sachs Global Investment Research, Visible Alpha Consensus Data

Onto Innovation - Financials

We outline our key assumptions for Onto Innovation's business over the next few years below. At a high level, we expect the company to outgrow the broader process control market growth, driven by strong momentum in both front end and advanced packaging for compute and memory due to tighter yield requirements in AI applications. We believe gross margins have room to expand in 2027 as the company ramps its Asia-based manufacturing footprint, and the company is poised to drive further OpEx leverage over the coming years.

Revenue: We forecast a 3-year revenue CAGR (CY2025-28) of ~35%, driven primarily by substantial growth in the Advanced Nodes (~36% CAGR) and Advanced Packaging segments (~56% CAGR) with a more measured pace in the Software and Services. We note that the Logic sub-segment rebounded in 2027 following a significant decline following market share loss at TSMC - and we would expect that this segment continues to rebound as Onto regains share.

Gross margin: We model 60 bps of margin expansion in 2026, mostly due to manufacturing efficiencies and improving next-generation product mix, followed by 140 bps of gross margin expansion in 2027. We believe Onto can increase gross margins in the long run as it continues to shift to its Asia-based manufacturing facility and shifts to its next-generation tools with higher ASPs.

Operating margin: We estimate 2026/27/28 operating margins of 30.7%/35.4%/35.3%, respectively. Consistent with management commentary, we expect R&D spending to see relatively less leverage, with greater SG&A leverage as the company continues to ramp its next-generation platforms.

EPS: We forecast 2025/26/27 operating EPS of \$8.25/\$12.50/\$15.00, respectively.

Price target: We have a 12-month price target of \$400, which is based on 32X our normalized EPS estimate of \$12.50. Our target P/E is in line with peer multiples, and it is on the higher end of ONTO's historical median, given recent momentum around HPC.

Key downside/upside risks:

- If Onto secures additional volume purchase agreements and accelerates customer adoption of its Dragonfly G5 platform, it would provide incremental revenue upside and stronger forward visibility.
- If advanced packaging capacity expansion shifts heavily toward leading-edge fabs where Onto and KLA maintain dominant positions, it could drive upside; conversely, a shift toward OSATs or domestic Chinese manufacturers could favor Camtek.
- If a competitor launches a rival tool with comparable throughput and cost of ownership, it could limit Onto's projected market share gains and pressure its top-line growth trajectory.
- If there is a broader slowdown in AI-related capital expenditure, Onto could face severe headwinds given its high exposure to AI-related advanced packaging and front-end wafer inspection.

Exhibit 11: Financial summary - ONTO

\$ mn except per share data

		2026 (E)					2027 (E)					2028 (E)				
	FY25 (A) 2025	1Q26 (A) Mar Q1	2Q26 (A) Jun Q2	3Q26 (E) Sep Q3	4Q26 (E) Jan Q4	FY26 (E) 2026	1Q27 (E) Mar Q1	2Q27 (E) Jun Q2	3Q27 (E) Sep Q3E	4Q27 (E) Dec Q4E	FY27 (E) 2027	1Q28 (E) Mar Q1E	2Q28 (E) Jun Q2E	3Q28 (E) Sep Q3E	4Q28 (E) Dec Q4E	FY28 (E) 2028
Total Revenue	1,005.3	291.9	343.1	398.2	420.0	1,453.3	447.1	483.6	523.8	554.6	2,009.1	581.3	607.2	628.4	650.5	2,467.3
% yoy	1.8%	9.5%	35.3%	82.5%	57.4%	44.6%	53.1%	40.9%	31.5%	32.0%	38.2%	30.0%	25.5%	20.0%	17.3%	22.8%
% qoq	--	9.4%	17.5%	16.1%	5.5%		6.5%	8.2%	8.3%	5.9%		4.8%	4.4%	3.5%	3.5%	
Cost of Revenue	505.5	145.6	159.9	178.5	183.8	667.8	188.5	202.1	217.2	228.7	836.4	239.0	248.9	257.1	265.7	1,010.8
Gross Profit	499.8	146.4	183.3	219.7	236.2	785.6	258.6	281.5	306.7	325.9	1,172.7	342.3	358.2	371.2	384.8	1,456.6
Gross Margin %	49.7%	50.1%	53.4%	55.2%	56.2%	54.1%	57.8%	58.2%	58.5%	58.8%	58.4%	58.9%	59.0%	59.1%	59.2%	59.0%
Operating Expenses:																
Research and Development	132.0	35.1	38.9	42.0	44.5	160.5	47.2	50.0	53.0	56.2	206.4	59.6	63.1	66.9	70.9	260.6
% of revenue	13.1%	12.0%	11.3%	10.5%	10.6%	11.0%	10.6%	10.3%	10.1%	10.1%	10.3%	10.2%	10.4%	10.7%	10.9%	10.6%
Sales and Marketing	70.0	21.5	23.1	25.0	26.2	95.7	27.8	29.4	31.2	33.1	121.5	34.4	35.8	37.2	38.7	146.1
% of revenue	7.0%	7.4%	6.7%	6.3%	6.2%	6.6%	6.2%	6.1%	6.0%	6.0%	6.0%	5.9%	5.9%	5.9%	5.9%	5.9%
General and Administrative	108.2	31.4	34.3	36.0	37.1	138.7	38.9	40.9	42.9	45.0	167.7	46.8	48.7	50.7	52.7	198.9
% of revenue	10.8%	10.8%	10.0%	9.0%	8.8%	9.5%	8.7%	8.4%	8.2%	8.1%	8.3%	8.1%	8.0%	8.1%	8.1%	8.1%
Operating Income	189.6	58.4	87.0	116.8	128.4	390.7	144.8	161.2	179.6	191.6	677.2	201.5	210.6	216.4	222.5	851.0
Operating Margin %	18.9%	20.0%	25.4%	29.3%	30.6%	26.9%	32.4%	33.3%	34.3%	34.5%	33.7%	34.7%	34.7%	34.4%	34.2%	34.5%
Net Income	136.8	33.8	60.1	84.6	94.3	272.7	109.4	122.3	136.8	146.0	514.5	152.2	159.9	164.9	170.0	646.9
Net Margin %	13.6%	11.6%	17.5%	21.2%	22.4%	18.8%	24.5%	25.3%	26.1%	26.3%	25.6%	26.2%	26.3%	26.2%	26.1%	26.2%
EPS - non-GAAP	\$4.94	\$1.42	\$1.93	\$2.38	\$2.53	\$8.25	\$2.76	\$3.01	\$3.28	\$3.46	\$12.50	\$3.57	\$3.71	\$3.81	\$3.90	\$15.00
% yoy						67.0%	94.8%	55.6%	38.3%	36.7%	51.5%	29.5%	23.6%	16.0%	12.9%	19.9%

Source: Company data, Goldman Sachs Global Investment Research

Disclosure Appendix

Reg AC

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